

## Group Participation Sheet

*Prof. Mouhamed Abdulla, School of Applied Computing, FAST  
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A portion of this course is devoted to group work. This **Group Participation Sheet (GPS)**, hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of “**zero**” for the mentioned assessment. Any concerns regarding this matter should be brought to the professor’s attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

|                                 |                                              |                                                                                                             |                             |
|---------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------|
| Course Title:                   |                                              |                                                                                                             |                             |
| Evaluation Name:                |                                              |                                                                                                             |                             |
| Team Number:                    | N/A                                          |                                                                                                             |                             |
| Referencing and Citation Style: | APA Style<br><input type="text"/>            | IEEE Style<br><input type="text"/>                                                                          | N/A<br><input type="text"/> |
| Group Members:                  | First Name and Family Name of group members: | Signature of group members:                                                                                 |                             |
|                                 | (1) (team leader)                            | Afolabi Adesina<br><small>Digitally signed by Afolabi Adesina<br/>Date: 2026.06.12 14:08:34 -04'00'</small> |                             |
|                                 | (2)                                          | Yanan Yang<br><small>Digitally signed by Yanan Yang<br/>Date: 2026.06.12 14:12:05 -04'00'</small>           |                             |
|                                 | (3)                                          | Xian Qin<br><small>Digitally signed by Xian Qin<br/>Date: 2026.06.12 14:18:02 -04'00'</small>               |                             |
|                                 | (4)                                          |                        |                             |
| (5)                             |                                              |                                                                                                             |                             |
| Submission Date:                |                                              |                                                                                                             |                             |

|                  |
|------------------|
| <b>Grade:</b> /5 |
|------------------|

## **AI/ML Capstone Project**

### **PR1 to PR7 Weekly Progress Report (5% of final mark)**

**Due: Week 6 to 11 (softcopy on SLATE, hardcopy to the Professor in class)**

|                                |                                    |
|--------------------------------|------------------------------------|
| <b>Name 1:</b> Afolabi Adesina | <b>Student Number 1:</b> 991808102 |
| <b>Name 2:</b> Evans Frimpong  | <b>Student Number 2:</b> 991771602 |
| <b>Name 3:</b> Xian Qin        | <b>Student Number 3:</b> 991578381 |
| <b>Name 4:</b> Yanan Yang      | <b>Student Number 4:</b> 991841037 |

#### **Instructions:**

- Please work on the following weekly progress report with the group.
- Prepare the deliverable report as discussed in class.
- Type the report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload the report in PDF format on SLATE. The team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stapled) of the report to the Professor in class.
- Any form of plagiarism and use of AI generated text in any part of the deliverable will result in a grade of zero for all members in the group. Teams involved in this form of academic misconduct will be reported for further disciplinary actions.

#### **Overall Objective:**

Over a 7 week period, students will engage in a sprint development cycle to deliver iterative progress with the AI/ML capstone project. Each week, the designated TL is responsible for submitting a report on behalf of the group. The dates for the working weeks are as follows:

| <b>Progress Reports</b> | <b>Working Week</b>                                   | <b>Submission Deadline</b> |
|-------------------------|-------------------------------------------------------|----------------------------|
| PR1 (5%)                | W5 and W6: Wed. Jun. 3, 2026 to Tue. Jun. 9, 2026     | W6: Fri. Jun. 12, 2026     |
| PR2 (5%)                | W6 and W7: Wed. Jun. 10, 2026 to Tue. Jun. 16, 2026   | W7: Wed. Jun. 17, 2026     |
| PR3 (5%)                | W7 and RW: Wed. Jun. 17, 2026 to Mon. Jun. 22, 2026   | RW: Mon. Jun. 22, 2026     |
| PR4 (5%)                | RW and W8: Tue. Jun. 23, 2026 to Tue. Jun. 30, 2026   | W8: Wed. Jul. 1, 2026      |
| PR5 (5%)                | W8 and W9: Wed. Jul. 1, 2026 to Tue. Jul. 7, 2026     | W9: Wed. Jul. 8, 2026      |
| PR6 (5%)                | W9 and W10: Wed. Jul. 8, 2026 to Tue. Jul. 14, 2026   | W10: Wed. Jul. 15, 2026    |
| PR7 (5%)                | W10 and W11: Wed. Jul. 15, 2026 to Tue. Jul. 21, 2026 | W11: Wed. Jul. 22, 2026    |

\* RW equals Reading Week

Instructor: Prof. Mouhamed Abdulla

The weekly PR report should address these sections. Use bulleted terms with the answers.

### Q1: Summary of Work

Provide a summary of the work completed during this current sprint. [1 mark]

- Found two additional datasets from UK and US that are useful to train that includes features that represent general location and attributes.
- Found one paper as reference to support us to pick up the features to train models, and which models use to do the experiments
- Draw the UML diagrams to help teams to refine the idea about the whole project

### Q2: List of Challenges

Provide a list of challenges encountered, and how the team addressed those challenges during this sprint. [1 mark]

- Hard to find the real-time camera image used to check the road conditions, however, we can get these info from 511API, even the values are not exactly same as the training dataset we use, we can preprocessing them to make them consistent
- To define how many coordinates were needed to use to fetch the weather and surface condition, we are still struggling to find the reference to support, the alternative workaround is based on the experiences to set up the initiate threshold values, and update it during experiments

### Q3: Technology Stack

Provide the current technology stack used with specific version numbers for tools, libraries, databases, and APIs. [1 mark]

- API: Google Map API, env\_canada, ECWeather, Ontario 511 API, `base_url="https://511on.ca/api/v2/get"`
- Library: Requests, Asyncio, *nest\_asyncio*.

### Q4: Meeting Minutes

Provide the details of the meeting minutes (those conducted in class and outside) and the GitHub activity. [1 mark]

- Review the UML diagram together and refine the project logic
- Decide the datasets and the features used to train.
- Create the GitHub and assign the access. Set priority to our goals and adjust the building strategy accordingly.

### Q5: Future Work

Provide a work plan for the upcoming week. [1 mark]

- Preprocessing datasets.
- Set up model experiments
- Search references used to support the algorithms, which is used to decide the threshold values to pick the coordinates, and decide the road surface by weather.

## References

References (Include potential references found and used to help with the AI/ML capstone project during this sprint).

- [1] J. Jiang, Y. Miao, and D. Wu, “Machine Learning based Prediction Analysis of Potential Factors in Traffic Accidents,” *Applied and Computational Engineering*, vol. 99, no. 1, pp. 112 to 120, Nov. 2024. [Online]. Available: <https://doi.org/10.54254/2755-2721/99/20251788>
- [2] C. Palay, “SDOT collisions all years [Data set],” City of Seattle ArcGIS Online, 2023. [Online]. Retrieved September 6, 2024, from <https://data-seattlecitygis.opendata.arcgis.com/maps/SeattleCityGIS::sdot-collisions-all-years-2>
- [3] Department for Transport, “Road safety data, collisions last 5 years [Data set],” 2025. [Online]. Retrieved September 6, 2024, from <https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-collision-last-5-years.csv>